Where Does Physics Live in Vision Encoders?

Spatially Probing and Amplifying Physical Reasoning in VLM Representations

Relevant Past Papers

1. **Pixels to Principles** (Ballout et al., arXiv 2507.16572, July 2025)
   Summary- It systematically evaluates 9 MLLMs across three families (InternVL 2.5, LLaVA-OneVision, Qwen 2.5 VL) on intuitive physics tasks from GRASP and IntPhys 2, finding that vision encoders capture physics plausibility cues at up to 85% probing accuracy (InternVL 2.5 78B) while end-to-end model output never exceeds 54%. The paper uses mean-pooled global embeddings with binary plausibility labels and t-SNE visualization showing physics concept clusters that dissolve after projection.
   Gap: The probing is binary (plausible vs. implausible) and globally pooled, it cannot identify which image patches encode specific physical properties or how continuous physics variables degrade spatially through the pipeline.

2. **Lost in Embeddings** (Li et al., EMNLP 2025 Findings, arXiv 2509.11986)
   Summary - It introduces two methods to quantify information loss at the VLM connector: k-Nearest Neighbor Overlap Ratio (KNOR) showing 40–60% geometric distortion post-projection, and patch-level reconstruction loss maps that localize where information is destroyed on the image.
   Gap: Their patch-level reconstruction loss overlays are applied to general VQA tasks (VizWiz, SeedBench), never to physics scenes, and never decomposed by physics variable type (mass, friction, stability).

3. **Hidden in Plain Sight** (Fu et al., COLM 2025 Outstanding Paper, arXiv 2506.08008)
   Summary- It performs the only controlled component ablation in VLMs, applying equal-parameter LoRA (16.7M params) to ViT, projector, and LLM independently across vision-centric tasks, finding that LLM fine-tuning provides the most substantial improvements while vision representations remain intact through the pipeline.
   Gap: The ablation covers depth estimation, correspondence, and art style , not physics reasoning. The finding that the LLM is the bottleneck for general vision tasks may not hold for physics, where the visual distribution shift from web images to physics scenarios is substantial.

4. **Physics Context Builders** (Balazadeh et al., ICCV 2025, arXiv 2412.08619)
   Summary - It fine-tunes PaliGemma-3B with LoRA across all modules on simulation-generated physics QA data, achieving up to 13.8% improvement on physical reasoning tasks and demonstrating sim-to-real transfer. Their partial ablation shows joint training outperforms

LM-only fine-tuning.
Gap: No isolated encoder-only ablation, no visualization of what the encoder learns, and their approach bypasses the projection bottleneck by converting physics to text via a separate small VLM rather than fixing the encoder's representations.

5. **PhyCritic** (Xiong et al., NVIDIA, arXiv 2602.11124, February 2026)
Summary- It demonstrates that physics knowledge already exists in VLM representations and can be surfaced through two-stage GRPO on Qwen2.5-VL-7B: 80 warmup steps on Cosmos-Reason1-RL data followed by 300 steps with a self-referential critic cycle using only 3,258 samples, achieving +12 points over all 7B baselines.
Gap: Full end-to-end training with no component isolation, impossible to determine whether improvements come from the ViT, projection, or LLM. No visualization of what changes in the representations.

6. **VideoREPA** (Zhang et al., NeurIPS 2025, arXiv 2505.23656)
Summary- Discovers that T2V diffusion model representations have weaker physics understanding than self-supervised video encoders (VideoMAEv2), and proposes Token Relation Distillation to align pairwise token similarities, improving physics commonsense by 24.1% on VideoPhy.
Gap: Operates on T2V diffusion models, not VLMs. Measures physics understanding via aggregate probing accuracy, never visualizing spatially where physics information lives in the token space.

7. **NeRF2Physics** (Zhai et al., CVPR 2024)
Summary- produces spatial 3D visualizations of predicted mass density, friction, and hardness on object surfaces by fusing CLIP features into neural radiance fields and using LLM-based material reasoning with zero-shot kernel regression.
Gap: Uses standalone CLIP features outside the VLM pipeline. Visualizes predicted outputs, not internal encoder representations. Never examines how physics features survive or degrade through the projection layer.

8. **VLM4VLA** (arXiv 2601.03309, January 2026)
Summary - Tests Qwen2.5VL-3B, 7B, and PaliGemma-1 on robotic manipulation, finding that freezing the vision encoder causes catastrophic performance drops of 24–42%, in stark contrast to **Hidden in Plain Sight**'s finding that LLM tuning matters most for general vision tasks.
Gap: Tests manipulation tasks, not physics reasoning. No visualization of encoder representations.

9. **From Diagnosis to Improvement: Probing Spatio-Physical Reasoning in VLMs** (arXiv 2508.10770, 2025)
Summary - Provides diagnostic analysis of mainstream VLMs on spatio-physical reasoning, revealing underperformance attributable to human-like priors and lack of deep reasoning, then applies SFT followed by rule-based RL to Qwen2.5-VL-7B for improvement.
Gap: Focuses on behavioral analysis (CoT error taxonomy) without probing internal representations or visualizing what physics features the encoder encodes.

Motivation

What limitation or problem are we solving and how do we know it exists?

Vision-Language Models form the backbone of every major Physical AI system deployed or announced in 2026, NVIDIA's Cosmos Reason 2 and GR00T N1.7, Google DeepMind's Gemini Robotics 1.5, and Physical Intelligence's π0.5 all build on VLM backbones to translate visual perception into physical reasoning and robotic action. Yet a convergent finding across multiple independent studies reveals a paradox: VLM vision encoders capture rich physics-relevant signals that are systematically lost before they can be used for reasoning. Pixels to Principles (Ballout et al., 2025) showed 85% probing accuracy on physics plausibility before the projection layer, collapsing to approximately 54% at the model's output. Lost in Embeddings (Li et al., 2025) quantified 40–60% geometric distortion of visual representations at the projection step. PhyCritic (Xiong et al., 2026) demonstrated that physics knowledge already exists in VLM representations and can be partially surfaced through training alone.

Despite this converging diagnosis, two critical questions remain unanswered. First, nobody has shown what physics "looks like" spatially within encoder representations. All existing probing studies use mean-pooled global embeddings (one number per video) or aggregate accuracy metrics. We do not know which image patches encode mass, which encode contact dynamics, which encode surface friction, or whether physics is encoded as a spatially structured object-level property or a globally diffused scene-level signal. Second, nobody has performed a complete factorial component ablation (encoder vs. projection vs. LLM) specifically for physics reasoning tasks. Hidden in Plain Sight found LLM fine-tuning is most impactful for general vision tasks, but VLM4VLA (2026) found the exact opposite for robotic manipulation. Physics reasoning likely falls between these extremes, but the answer is unknown.

Why is this limitation important?

Every frontier company building Physical AI treats the VLM backbone as an opaque black box. NVIDIA's strategy at GTC 2026 is to "turn robotics' data problem into a compute problem" through simulation and synthetic data, but PhysBench (ICLR 2025) showed that more data does not fix physics reasoning failures. Without understanding where physics information lives, where it degrades, and which component to target for improvement, the field is applying expensive interventions (full-model GRPO, end-to-end SFT, custom encoder architectures) without knowing whether they are addressing the right bottleneck.

Why does our idea solve it?

We propose the first spatial physics probing study combined with a targeted component ablation for physics reasoning. By training patch-level probes on VLM encoder representations using simulation datasets with ground-truth continuous physics variables, we produce "physics saliency maps", heatmaps overlaid on images showing which patches encode mass, friction, and stability. By comparing these maps before and after the projection layer, we reveal exactly where in the image physics information is lost. By then applying LoRA to individual VLM components (encoder, projection, LLM) and regenerating the physics saliency maps, we determine which component is the most impactful target for physics reasoning and show mechanistically how fine-tuning changes the spatial encoding of physics.

Why would our idea probably work?

Three independent lines of evidence support feasibility. First, Lost in Embeddings already demonstrated that patch-level information loss can be localized on images, we extend this methodology from general VQA to physics using simulation-grounded ground truth. Second, Pixels to Principles confirmed that binary physics probing works well before projection (85% accuracy), extending from binary to spatial-continuous probing is a methodological step, not an architectural leap. Third, the component ablation methodology is validated by Hidden in Plain Sight and VLM4VLA (2026), we apply the same approach to the untested physics domain using established tools (LoRA, PEFT library) on readily available open-source models and datasets.

Key Ideas/Contributions/Novelty

Contribution 1: Physics Saliency Maps , the first spatial visualization of physics encoding in VLM representations.

No existing work has produced a visualization showing which image patches in a VLM's vision encoder activate for specific physical properties. We will generate patch-level physics probing heatmaps for mass, friction, elasticity, and stability, revealing whether physics is encoded as spatially localized object-level features (contact points, material surfaces, object volumes) or globally diffused scene-level signals. These maps will be generated at multiple pipeline stages (pre-projection, post-projection, early LLM layers) to show where the spatial physics signal degrades, extending Lost in Embeddings' general information loss maps to the physics domain. This contribution produces immediately compelling visual results and provides mechanistic insight no prior paper offers.

Contribution 2: The first factorial component ablation for physics reasoning in VLMs.

We will apply equal-parameter LoRA (following Hidden in Plain Sight's methodology) to the vision encoder alone, the projection layer alone, and the LLM's early layers alone, evaluated on physics-specific benchmarks (PhysBench, GRASP Level 2, Physion++). This directly answers whether physics reasoning follows the "LLM is the bottleneck" pattern (Fu et al., COLM 2025) or the "encoder is critical" pattern (VLM4VLA, 2026), resolving a genuine empirical contradiction in the literature. No prior work has tested this for physics.

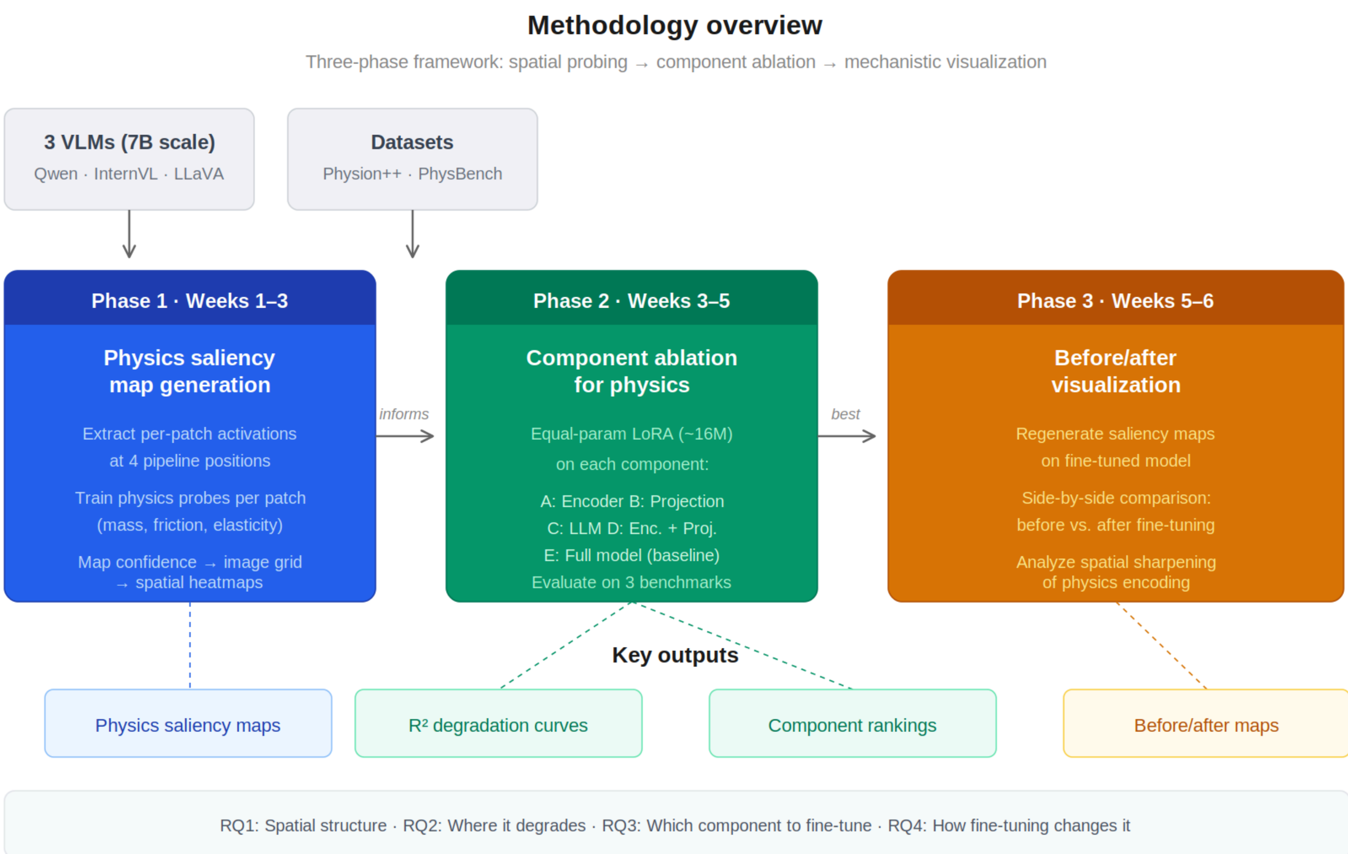
Contribution 3: Before/after visualization showing how targeted fine-tuning changes physics spatial encoding.

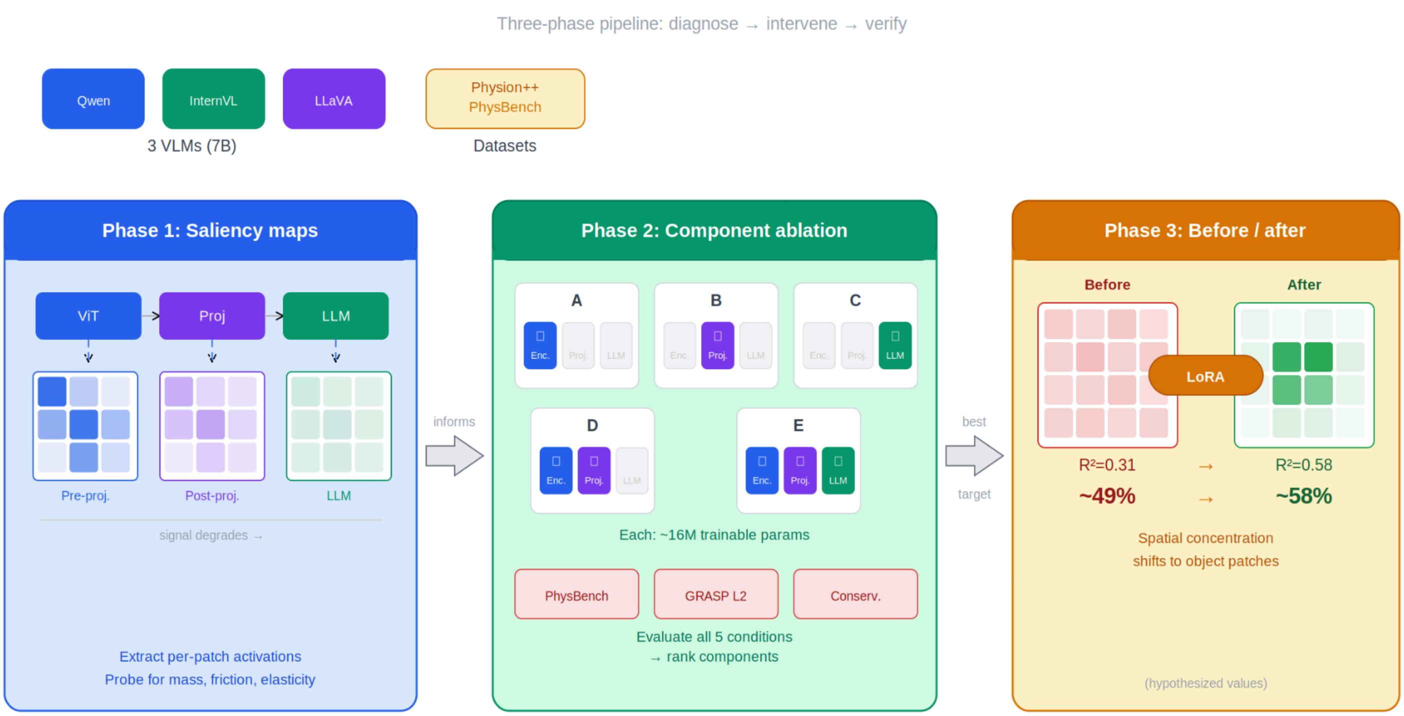
After the component ablation identifies the most impactful target, we regenerate physics saliency maps to show mechanistically what fine-tuning changes in the encoder's internal representations. This connects the diagnostic (Contribution 1) to the intervention (Contribution 2) into a coherent narrative: "here is where physics lives, here is where it dies, here is what targeted fine-tuning does to recover it." No existing physics fine-tuning paper (PhyCritic, PCBs, Cosmos-Reason1) has provided this kind of mechanistic visualization.

We therefore, are aiming to address the following research questions through this research:

1. What does physics look like spatially in VLM encoder representations, is it localized to physically relevant patches or globally diffused?
2. Where does the spatial physics signal degrade through the encoder to projection to LLM pipeline?
3. For physics reasoning specifically, is the vision encoder, the projection layer, or the LLM the most impactful fine-tuning target?
4. How does targeted fine-tuning change the spatial encoding of physics in the encoder's representations?

Methods

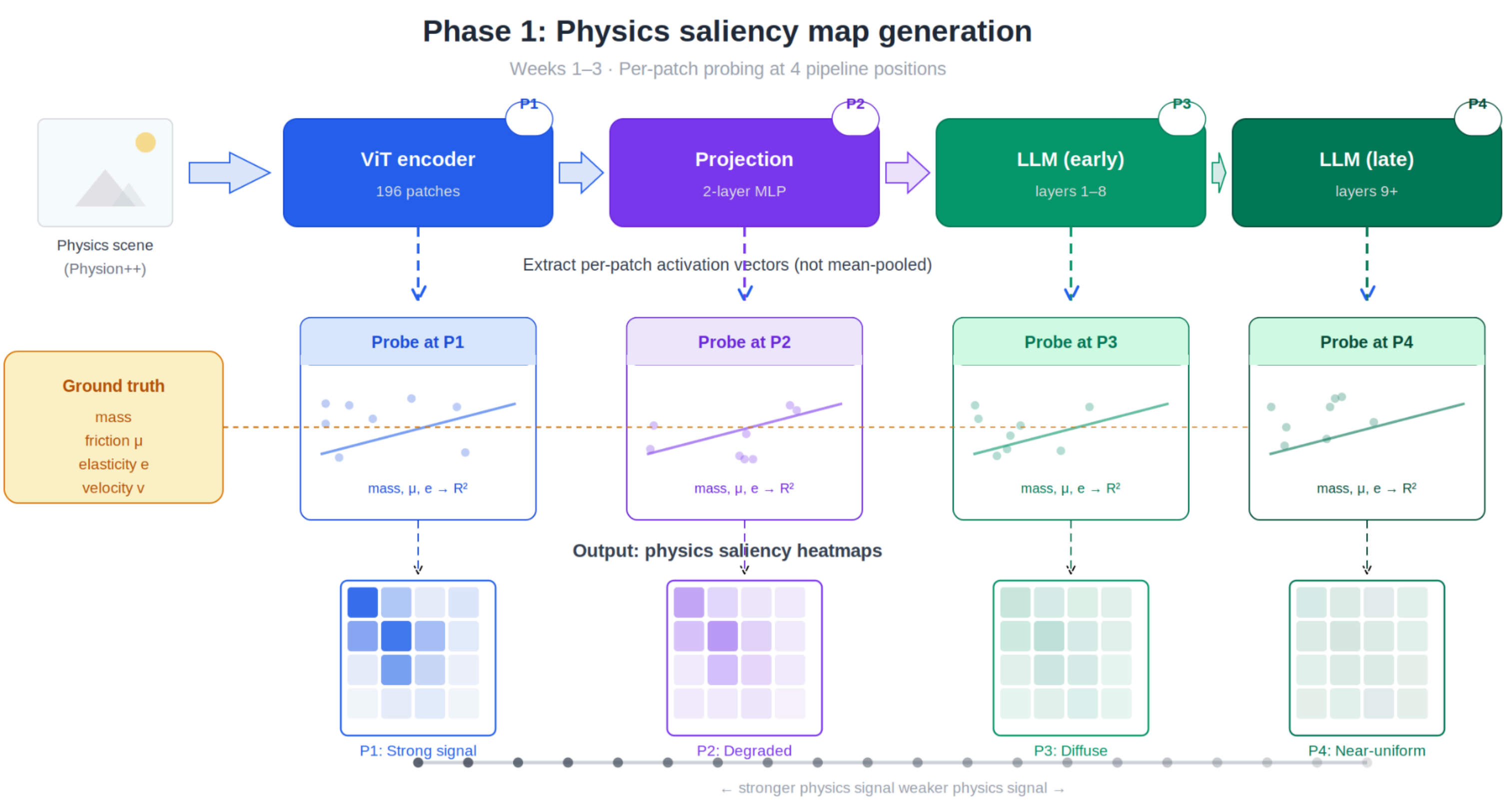

Methodology overview
Three-phase framework: spatial probing → component ablation → mechanistic visualization
3 VLMs (7B scale)
Qwen · InternVL · LLaVA
Datasets
Physion++ · PhysBench
Phase 1 · Weeks 1–3
Physics saliency map generation
Extract per-patch activations at 4 pipeline positions
Train physics probes per patch (mass, friction, elasticity)
Map confidence → image grid → spatial heatmaps
informs
Phase 2 · Weeks 3–5
Component ablation for physics
Equal-param LoRA (~16M) on each component:
A: Encoder B: Projection
C: LLM D: Enc. + Proj.
E: Full model (baseline)
Evaluate on 3 benchmarks
best
Phase 3 · Weeks 5–6
Before/after visualization
Regenerate saliency maps on fine-tuned model
Side-by-side comparison: before vs. after fine-tuning
Analyze spatial sharpening of physics encoding
Key outputs
Physics saliency maps
R² degradation curves
Component rankings
Before/after maps
RQ1: Spatial structure · RQ2: Where it degrades · RQ3: Which component to fine-tune · RQ4: How fine-tuning changes it


Three-phase pipeline: diagnose → intervene → verify
Qwen
InternVL
LLaVA
3 VLMs (7B)
Physion++
PhysBench
Datasets
Phase 1: Saliency maps
ViT
Proj
LLM
Pre-proj.
Post-proj.
LLM
signal degrades →
Extract per-patch activations
Probe for mass, friction, elasticity
informs
Phase 2: Component ablation
A
B
C
D
E
Enc.
Proj.
LLM
Each: ~16M trainable params
PhysBench
GRASP L2
Conserv.
Evaluate all 5 conditions
→ rank components
best
target
Phase 3: Before / after
Before
After
LoRA
R²=0.31
R²=0.58
~49%
~58%
Spatial concentration
shifts to object patches
(hypothesized values)

Phase 1: Physics Saliency Map Generation

We extract per-patch activation vectors (not mean-pooled) from VLM vision encoders at four pipeline positions: (1) vision encoder output (pre-projection), (2) post-projection, (3) LLM layer 8, and (4) LLM layer 16. For each patch position, we train lightweight probes (linear ridge regression and single-hidden-layer MLP with 512 units) to predict continuous physics variables from that patch's activation vector. Ground truth comes from Physion++ simulation metadata, which provides per-object mass, dynamic friction, static friction, elasticity, and per-frame positions/velocities/angular velocities.

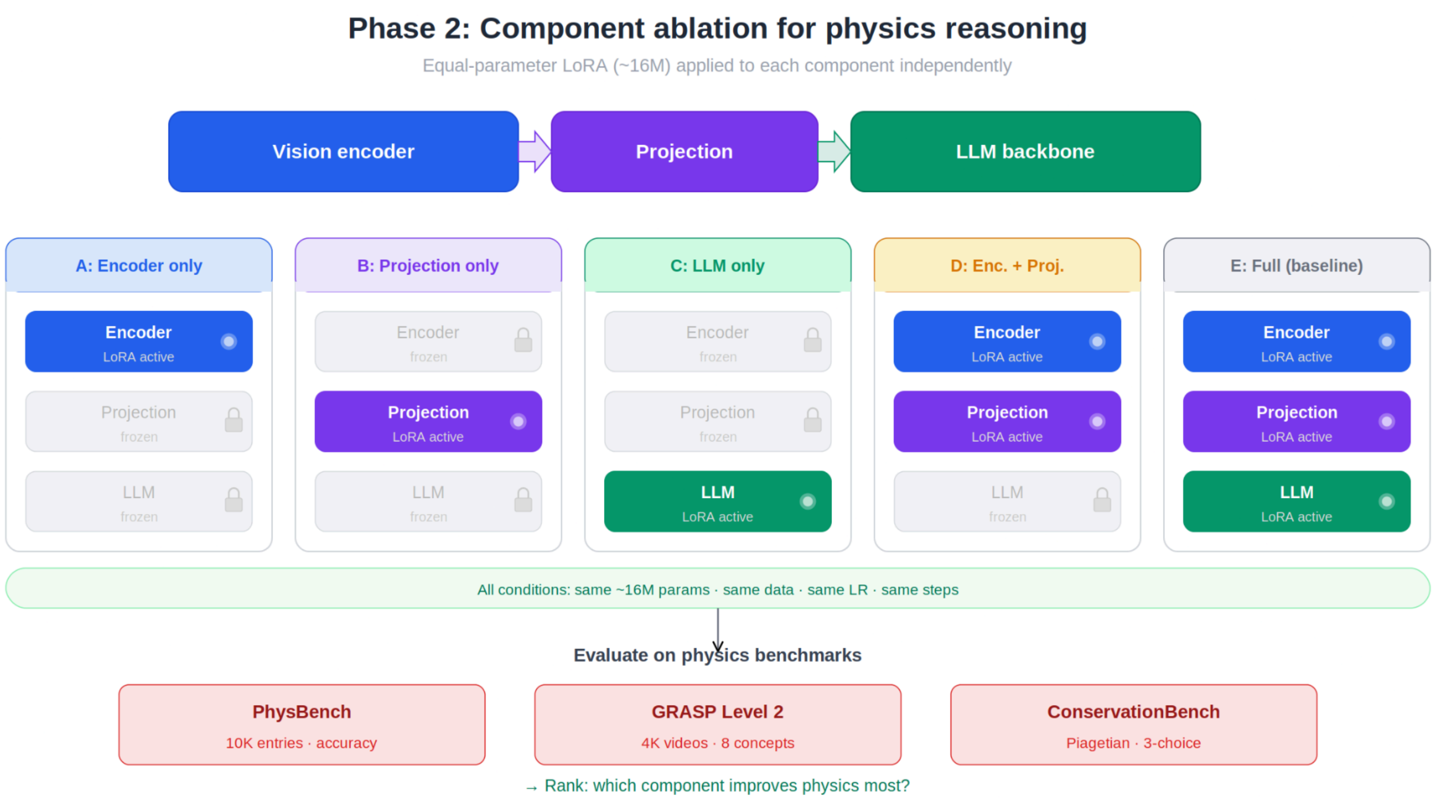
The probe's prediction confidence (for classification) or R-squared (for regression) at each patch location is then mapped back onto the original image grid (14x14 for a 224x224 input with 16x16 patches), upscaled via bilinear interpolation, and overlaid as a heatmap on the original image. This produces "physics saliency maps", spatial visualizations showing which image regions encode mass, friction, elasticity, and contact dynamics at each pipeline stage.



For physics variables that are object-level (mass, friction coefficient), we assign ground-truth labels to patches based on their spatial overlap with object segmentation masks from the simulation renderer. For scene-level dynamics (stability, collision outcome), we use temporal derivatives of position/velocity to compute per-patch physics activity scores.

We run this on three open-source VLM families at 7B scale: Qwen2.5-VL-7B-Instruct (unfrozen ViT during pretraining), InternVL 2.5-8B (frozen ViT with separate incremental learning stage), and LLaVA-OneVision-7B (frozen ViT throughout). This comparison across ViT training strategies tests whether unfreezing the ViT during pretraining (Qwen) produces more physics-informative spatial representations than keeping it frozen (LLaVA).

Phase 2: Component Ablation for Physics



Following Hidden in Plain Sight's methodology, we apply LoRA with equal trainable parameter count (~16M) to each VLM component independently:

Condition A (Encoder-only): LoRA rank 16 on the vision encoder's later attention layers (Q/V projections in the last 6 ViT blocks). Vision encoder frozen during standard VLM pretraining is now unfrozen via LoRA.

Condition B (Projection-only): LoRA on the 2-layer MLP projector's linear layers. This is the smallest component but directly targets the bottleneck Pixels to Principles identified.

Condition C (LLM-only): LoRA rank 16 on the LLM's first 8 layers' Q/V attention projections. This follows the finding from Hidden in Plain Sight and Amazon's LoRA module study that o_proj and early LLM layers are the most productive targets.

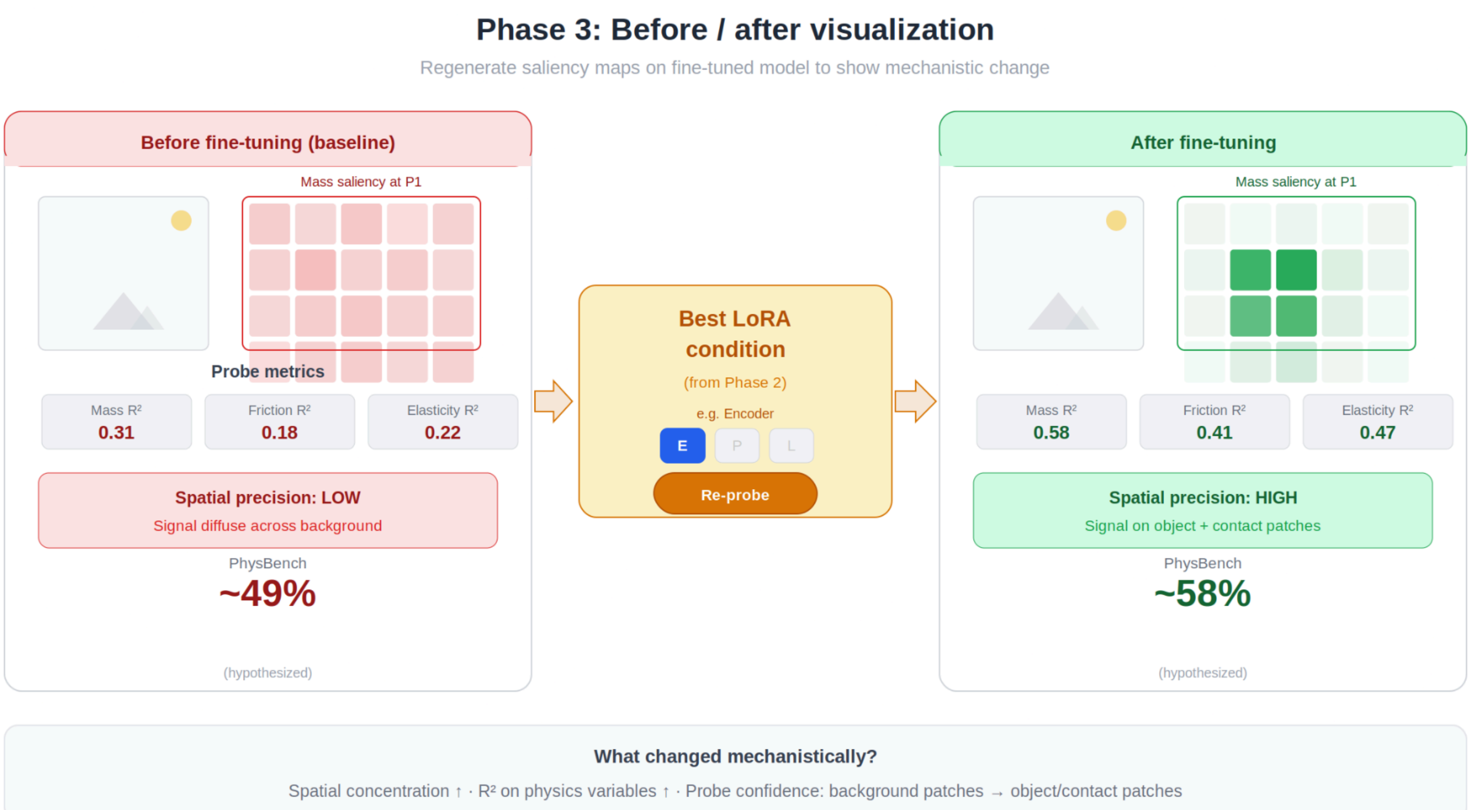
Condition D (Encoder + Projection): Combined LoRA on both encoder and projector, testing whether the projection bottleneck is best fixed by jointly improving both sides.

Condition E (Full model baseline): LoRA on all components, replicating the standard approach used by PCBs and Cosmos-Reason1.

All conditions are trained on the same physics-grounded data from Physion++ (rendered frames paired with physics QA derived from simulation metadata) and evaluated on PhysBench, GRASP Level 2, and ConservationBench.

Phase 3: Before/After Physics Saliency Visualization

After identifying the most impactful component from Phase 2, we regenerate the physics saliency maps from Phase 1 using the fine-tuned model. This produces side-by-side comparisons: "before fine-tuning" vs. "after fine-tuning" saliency maps for each physics variable on the same images. We analyze whether fine-tuning sharpens the spatial physics encoding (concentrating probe confidence on physically relevant patches like contact points and material surfaces), broadens it (improving all patches uniformly), or has no spatial effect (improving only the aggregate score without changing spatial structure).



Experimental Setup

Baselines for comparison:

- **Baseline 1 (No intervention):** Standard VLM inference on physics benchmarks, establishes the performance floor.
- **Baseline 2 (Text-mediated physics, PCB-style):** Provide the VLM with ground-truth physics descriptions as text context. This tests whether text-mediated grounding closes the gap without any model modification.
- **Baseline 3 (Self-referential physics, PhyCritic-style)**: Two-step prompting where the VLM first predicts which physics principles govern the scene, then uses its own answer as context. This tests whether the VLM's existing physics knowledge suffices when explicitly activated.
- **Baseline 4 (Full-model end-to-end fine-tuning)**: LoRA on all components, the standard approach, providing the performance ceiling for LoRA-based methods.

Ablation testing to reduce confounding factors:

Each component ablation condition uses the same total number of trainable parameters (~16M), same training data, same learning rate schedule, same number of training steps, and same evaluation protocol. This follows Hidden in Plain Sight's methodology to ensure differences are attributable to the component targeted, not to parameter count or training regime.

We additionally ablate the number of LoRA-adapted ViT layers (last 2 vs. last 6 vs. all), LoRA rank (8 vs. 16 vs. 32), and ViT learning rate relative to LLM learning rate (same vs. 5x lower vs. 10x lower, following LLaVA-OneVision's finding that differentiated learning rates matter).

Visualization and statistical analysis:

Physics saliency maps with heatmap overlays on physics scene images (the primary visual contribution). R-squared degradation curves across pipeline stages for each physics variable (mass, friction, elasticity). Paired comparison of saliency maps before and after fine-tuning. Per-physics-concept accuracy breakdown across component conditions. Attention weight analysis on physics vs. non-physics patches in the LLM layers.

Datasets and Evaluation

Training dataset:

Physion++ (NeurIPS 2023 Datasets & Benchmarks). Contains video plus per-object ground-truth mass, dynamic/static friction, elasticity, per-frame positions, velocities, angular velocities, and collision events across 9 scenarios (dominoes, collide, contain, drop, link, roll, support, drape, occlude). Downloadable via S3 with .pkl metadata. We generate physics QA pairs from the simulation metadata for fine-tuning.

Evaluation benchmarks:

- PhysBench : 10,002 entries across 4 dimensions of physical world understanding. The primary standardized benchmark, allows comparison against 75 published VLM results. Metric: accuracy.
- GRASP Level 2: 4,096 videos spanning 8 physics concepts (collision, gravity, continuity, object permanence, solidity, inertia, unchangeableness, gravity-support). Binary plausibility classification. Allows direct comparison with Pixels to Principles results. Metric: accuracy.
- ConservationBench: Piagetian conservation tasks testing whether models recognize that physical quantities remain invariant under spatial transformations. 3-choice format. Most VLMs cluster near 33.3% chance level. Metric: accuracy.
- Physion++ test set: Used for continuous physics probing evaluation. Metric: R-squared for regression probes on mass, friction, elasticity.

Evaluation metrics for physics saliency maps:

Spatial precision (fraction of high-confidence probe patches that overlap with ground-truth object masks), spatial recall (fraction of physics-relevant object patches captured by high-confidence probes), information loss ratio (R-squared drop from pre- to post-projection for each physics variable at each spatial location).

Benchmarks/Evaluation Sets

Baselines for comparison with existing systems:

1. GPT-4o on PhysBench: approximately 49.5% (Chow et al., 2025), the published state of the art at the time of PhysBench's release.
2. PhysAgent (GPT-4o + vision foundation models + physics knowledge memory): +18.4% improvement over base GPT-4o on PhysBench (Chow et al., 2025).
3. PhyCritic (Qwen2.5-VL-7B + two-stage GRPO): +12 points over all 7B baselines on PhyCritic-Bench (Xiong et al., 2026).
4. Cosmos-Reason1-7B (Qwen2.5-VL-7B + Physical AI SFT + GRPO): doubled baseline scores on physical common sense benchmarks (NVIDIA, 2025).
5. PCBs (PaliGemma-3B + simulation LoRA): +13.8% on Falling Tower stability detection (Balazadeh et al., 2025).
6. Gemini 2.0 Flash Thinking on GRASP: 53.7% (Ballout et al., 2025).
7. All 9 models from Pixels to Principles on GRASP Level 2: 35–54% range (Ballout et al., 2025).
8. 112 VLMs on ConservationBench: most cluster near 33.3% chance (Luo et al., 2025).

Ideal Results

Hypothesis 1 (Physics spatial encoding)

Physics information in VLM encoder representations is spatially structured, mass saliency concentrates on object volume patches, friction saliency concentrates on surface/contact patches, stability saliency concentrates on contact-point and center-of-mass patches. If confirmed, this demonstrates that the vision encoder has learned physically meaningful spatial representations, not just global scene-level heuristics.

Hypothesis 2 (Spatial degradation at projection)

Physics saliency maps become spatially diffuse after the projection layer, the concentrated physics-relevant patches become uniformly low-confidence across all patches. This would visually demonstrate the Lost in Embeddings finding specifically for physics, connecting their geometric distortion metric to spatially localized physics information loss.

Hypothesis 3 (Component ablation)

For physics reasoning, encoder fine-tuning will be more impactful than the Hidden in Plain Sight result suggests for general vision tasks, because physics scenes (simulation-rendered dynamics, contact mechanics, material interactions) represent a larger visual distribution shift from web-scale pretraining data than natural images do. We predict encoder + projection LoRA will outperform LLM-only LoRA on physics benchmarks by 5-15%, reversing the ranking Hidden in Plain Sight found for general vision.

Hypothesis 4 (Fine-tuning sharpens spatial physics)

After targeted encoder fine-tuning, physics saliency maps will show increased spatial concentration on physically relevant patches. Before/after comparison will show probe confidence shifting from diffuse background patches to concentrated object-level and contact-level patches, providing mechanistic evidence of what physics-grounded fine-tuning actually does to the representation.

Best-case scenario

Physics saliency maps reveal a clear, spatially structured encoding (Hypothesis 1 confirmed); the projection layer destroys this spatial structure (Hypothesis 2 confirmed); encoder fine-tuning is the most impactful intervention for physics (Hypothesis 3 confirmed); and fine-tuning visibly sharpens the spatial physics encoding in before/after visualizations (Hypothesis 4 confirmed). This would provide the first complete mechanistic picture, diagnosis, localization, intervention, and visual verification, of the physics reasoning bottleneck in VLMs.

Potential Limitations

Computation limits

Physics saliency map generation requires forward passes through 3 VLMs at 7B scale with per-patch activation extraction, which is memory-intensive (approximately 40GB GPU RAM per model for full activation caching). The component ablation requires 5 LoRA training runs per model, each approximately 6–12 hours on a single A100. Total estimated compute is approximately 950 A100-hours.

Generalization limits

Our physics probing uses Physion++ ground truth, which provides only four continuous physics variables (mass, friction, elasticity, and positional dynamics) across simulated scenes. Real-world physics involves many more variables (aerodynamics, fluid dynamics, deformable bodies) that our probing cannot capture. Additionally, simulation-rendered images differ visually from natural images, any fine-tuning improvements may reflect adaptation to the simulation visual domain rather than genuine physics understanding.

Dataset limits

Physion++ is built on ThreeDWorld with a specific visual rendering style. The physics saliency maps may not generalize to other simulation environments (MuJoCo, Isaac Sim) or real-world physics videos. PhysBench's physical reasoning tasks are primarily static image-based, while many physics phenomena are inherently dynamic and require video understanding.

Patch-level probing limitations

The 14x14 patch grid (for 224x224 input with 16x16 patches) provides coarse spatial resolution. Physics-relevant features that span less than one patch (e.g., a thin contact edge between objects) may be missed. Linear probes may underestimate nonlinear physics encoding; MLP probes address this but are harder to interpret spatially.

Scope limits

We restrict our study to three VLM families at 7B scale. Results may differ for larger models (70B+) or different architectures (MoE-based VLMs, SSM-based encoders). The component ablation uses LoRA, which approximates full fine-tuning but may not capture the full effect of unfreezing components.

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